WEEK 1: Networking Fundamentals | Days: 1-7

Day 2 Tasks— TCP/IP Basics

- **Task: ping google.com and observe the output.**

Window Cmd Output

```
Microsoft Windows [Version 10.0.19045.6937]
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C:\Users\SMI>ping www.google.com

Pinging www.google.com [142.251.151.119] with 32 bytes of data:
Reply from 142.251.151.119: bytes=32 time=880ms TTL=114
Reply from 142.251.151.119: bytes=32 time=296ms TTL=114
Reply from 142.251.151.119: bytes=32 time=595ms TTL=114
Reply from 142.251.151.119: bytes=32 time=467ms TTL=114

Ping statistics for 142.251.151.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 296ms, Maximum = 880ms, Average = 559ms

C:\Users\SMI>_
```

Kali Linux Terminal Output

- **TASK: Tracert (Windows) / traceroute (Linux) to any domain?**

```
┌──(root㉿kali)-[/home/kali/Desktop]
└─# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=907 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=1198 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=5455 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=4472 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=128 time=3464 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=128 time=2441 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=128 time=1417 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=128 time=393 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=128 time=2518 ms
```

Window Cmd Output

```
C:\Users\SMI>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1      3 ms      1 ms      1 ms  10.71.253.25
  2      *         *         *     Request timed out.
  3    190 ms    132 ms    262 ms  10.75.10.1
  4     68 ms     67 ms     75 ms  10.180.26.62
  5     50 ms     92 ms     62 ms  10.81.91.34
  6     96 ms     54 ms     65 ms  10.81.191.21
  7    119 ms     62 ms    105 ms  110.93.202.104
  8    111 ms     98 ms    124 ms  110.93.254.110
  9    197 ms     50 ms     66 ms  110.93.252.224
 10     94 ms     91 ms    353 ms  110.93.192.35
 11    113 ms    104 ms    117 ms  209.85.241.201
 12    117 ms    112 ms    104 ms  142.251.50.213
 13     94 ms     76 ms     90 ms  dns.google [8.8.8.8]

Trace complete.

C:\Users\SMI>
C:\Users\SMI>tracert 10.71.253.8

Tracing route to ASAD [10.71.253.8]
over a maximum of 30 hops:

  1     <1 ms     <1 ms     <1 ms  ASAD [10.71.253.8]

Trace complete.
```

Kali Linux terminal Output

```
┌──(root㉿kali)-[/home/kali/Desktop]
└─# traceroute google.com
traceroute to google.com (142.251.37.142), 30 hops max, 60 byte packets
 1  192.168.117.2 (192.168.117.2)  0.583 ms  0.256 ms  0.236 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * *
```

➢ TASK: Differences between TCP and UDP?

Answer:

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are two core protocols of the Transport Layer of the OSI and TCP/IP models. Both are responsible for end-to-end communication between applications, but they differ significantly in terms of reliability, speed, and use cases. Understanding the difference between TCP and UDP is essential for designing efficient and reliable networked systems. Protocols are rules that govern how data is formatted and sent over a network. TCP and UDP are two different methods for doing the same job: transferring data via the internet. They enable servers and devices to communicate so you can send emails, watch Netflix, play games, and browse web pages.

Transmission Control Protocol (TCP)

TCP is a reliable, connection-oriented transport protocol that ensures accurate and ordered data delivery. It uses control mechanisms to guarantee data correctness, which makes it slower but dependable.

- Connection-oriented protocol
- Reliable and ordered data delivery
- Higher overhead but high accuracy

User Datagram Protocol (UDP)

UDP is a fast, connectionless transport protocol that sends data without reliability guarantees. It is efficient for applications where speed is more important than accuracy.

- Connectionless and lightweight
- No guarantee of delivery or order
- Low overhead and high speed

Differences between TCP and UDP

TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection-oriented; uses a three-way handshake	Connectionless; no handshake
Guarantees reliable data delivery	Does not guarantee delivery
Uses acknowledgements (ACKs)	No acknowledgements
Supports retransmission of lost packets	No retransmission support
Ensures packets are delivered in order	Does not ensure ordering
Provides flow control and congestion control	No flow or congestion control
Slower due to higher overhead	Faster with minimal overhead
Variable header size (20–60 bytes)	Fixed header size (8 bytes)
Treats data as a continuous byte stream	Treats data as independent messages
Does not support broadcasting or multicasting	Supports broadcasting and multicasting
Used by HTTP, HTTPS, FTP, SMTP	Used by DNS, DHCP, VoIP, Streaming

Which protocol is better: TCP or UDP?

It depends on what you're doing online and the type of data being transferred. UDP is better if you're gaming online, because its speedy data transfer allows for mostly lag free gaming. TCP is better if you're transferring files, like family photos, because it ensures the data arrives exactly as it was sent. Overall, TCP and UDP are both useful protocols, so to think in terms of TCP vs UDP is a bit misleading. But depending on the type of data transfer, TCP or UDP might be better for the job. Here are some examples:

TCP is best for:

Email or texting
File transfers
Web browsing

UDP is best for:

live streaming
online gaming
Video chat

➢ TASK:Layer TCP/IP model vs 7-layer OSI model.

The 7-layer OSI model is a theoretical, vendor-neutral framework for understanding network communication, while the 4-layer TCP/IP model is the practical, protocol-specific implementation that powers the internet. OSI offers detailed separation of layers (Application-Physical), whereas TCP/IP combines similar functions (e.g., Application/Presentation/Session combined) for efficiency.

Key Comparisons:

- **Structure:** OSI has 7 layers (Physical, Data Link, Network, Transport, Session, Presentation, and Application); TCP/IP has 4 layers (Network Access, Internet, Transport, and Application).
- **Application & Use:** OSI is mainly used for teaching/reference, while TCP/IP is the actual protocol suite used for modern networking.
- **Layer Consolidation:**
  - OSI Layers 5-7 (Session, Presentation, and Application) are merged into the TCP/IP Application Layer.
  - OSI Layers 1-2 (Physical, Data Link) are merged into the TCP/IP Network Access Layer.
- **Reliability:** TCP/IP model supports only connectionless communication at the Network Layer, but both connection-oriented and connectionless at the Transport Layer. The OSI model supports both at the Network Layer.

Comparison Table:

Layer #	OSI Model	TCP/IP Model	Main Responsibility
7	Application	**Application**	Network apps (HTTP, FTP, DNS)
6	Presentation	**Application**	Encryption, formatting
5	Session	**Application**	Managing sessions

4	Transport	**Transport**	End-to-end connections (TCP/UDP)
3	Network	**Internet**	Routing, logical addressing (IP)
2	Data Link	**Network Access**	MAC addressing, framing
1	Physical	**Network Access**	Raw transmission (cables, bits)

Why the Differences Matter:

- **Troubleshooting:** The 7-layer OSI model is better for isolating problems because its layers are more specific.
- **Implementation:** TCP/IP is the protocol suite designed before the model itself, making it the practical choice for building networks.

Key Differences

Aspect	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Development	International standard (ISO)	Practical implementation (DoD)
Approach	Theoretical, "what to do"	Practical, "how it works"
Protocols	Protocol-independent	built around TCP/IP suite
Usage	Teaching, documentation	Actual internet implementation

Which One Matters?

TCP/IP Used on the actual internet. What engineers work with daily?

OSI Useful for conceptual understanding, troubleshooting, and certification exams.

OSI Layers (Bottom to Top).

Physical, Data Link, Network, Transport, Session, Presentation, Application.

TCP/IP Layers:

- All Internet Traffic Needs (Application, Internet, Transport, Network Access).

When to Use Each

Scenario	Use OSI	Use TCP/IP
• Real-world networking	✗	✓
• Studying for exams	✓	✓
• Troubleshooting conceptually	✓	✗
• Configuring routers	✗	✓
• Learning network fundamentals	✓	✗

The OSI model is great for understanding concepts, but TCP/IP is what actually powers the internet.